

Amelia Eckard

Charlotte, NC | ameliaeckard.com | github.com/ameliaeckard | linkedin.com/in/ameliaeckard

Education

University of North Carolina at Charlotte Charlotte, NC
M.S. in Artificial Intelligence / Early Entry Program *Expected May 2028*

University of North Carolina at Charlotte Charlotte, NC
B.S. in Computer Science / Concentration: AI, Robotics, & Gaming *Expected May 2027*

Experience

Research Intern Remote
Stevens Institute of Technology, Dr. Jina Huh-Yoo *June 2026 – Present*

- Engineered *rq-system*/LEAF, an AI-assisted literature-analysis pipeline using PubMed E-utilities, CrossRef, and LLM APIs to screen, classify, and extract structured evidence from 345 papers; processed 256 included, 80 excluded, and 9 review-flagged papers.
- Implemented dataset tagging, a seven-category research-question taxonomy, explicit screening/classification/extraction gates, and gold-sample validation using precision, recall, F1, accuracy, and confusion matrices.

Undergraduate Researcher Charlotte, NC
UNC Charlotte, Dr. Todd Dobbs *August 2025 – May 2026*

- Developed an Apple Vision Pro indoor-navigation prototype for visually impaired users using Swift, ARKit, RealityKit, ObjectTrackingProvider, Core ML, HRTF spatial audio, and distance-based pitch cues.
- Investigated object tracking, on-device recognition, and spatial-audio guidance as complementary modalities for accessible indoor navigation. Designed evaluation protocols for object identification, tracking reliability, navigation performance, and spatial-audio feedback workflows.

Lead Instructional Assistant, ITSC 1213 Charlotte, NC
UNC Charlotte College of Computing and Informatics *August 2026 – Present*

- Serve as lead instructional assistant for an asynchronous Python programming course, coordinating grading, rubrics, office hours, student support, and course logistics with the instructor.

Projects

CNN Scene Classification | *Python, PyTorch, torchvision, CUDA* 2026

- Trained and evaluated convolutional neural networks for 16-class scene recognition on 2,400 images.
- Improved validation accuracy from 42.25% to 51.46% through image augmentation while comparing preprocessing choices, training behavior, and model performance.

Emergency Fund Predictor | *Python, scikit-learn, pandas, SHAP* 2025

- Built an 11-feature machine-learning pipeline using 12,295 Federal Reserve SHED respondents enriched with BEA indicators to study emergency-fund vulnerability.
- Compared three classifiers and applied SHAP explainability to identify the strongest drivers of financial fragility and model predictions.

Resolve – LPL Financial University Hackathon | *AI/ML, Workflow Intelligence, FinTech* 2026

- Developing a proactive exception-intelligence layer that predicts likely blockers before financial-service requests visibly stall.

Vision Lab: ASCII Reconstruction | *JavaScript, HTML/CSS, Image Processing* 2026 – Present

- Developed an image-to-ASCII reconstruction experiment for a web-based computer-vision lab connected to ameliaeckard.com.

Technical Skills

Languages Python, SQL, Java, JavaScript, Swift

AI/Data PyTorch, torchvision, scikit-learn, NumPy, pandas, OpenCV, SHAP, Core ML, Ollama

Tools/APIs CUDA, Git, Docker, Railway, PubMed E-utilities, CrossRef, Perplexity API

Certifications & Involvement

Kode With Klossy × Goldman Sachs Machine Learning Challenge December 2025

UR2PhD Training Course, Computing Research Association December 2025